

# VISHAAL S

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Actively learning, growing and seeking more challenges to expand my knowledge and experience so that I can positively impact and excel in any environment I am put into.

## SKILLS

|                      |                     |                             |                |
|----------------------|---------------------|-----------------------------|----------------|
| DataBricks (PySpark) | PowerBI             | SQL                         | Pandas & Numpy |
| LLM & RAG Systems    | Agentic AI (CrewAI) | FastAPI (level 3 - HATEOAS) | PyTorch        |

## EXPERIENCE

### WALMART CENTER OF TECH EXCELLENCE - AI Engineer Intern

Jun 2025 - Aug 2025

- Worked on a project to identify emissions of harmful gases from micro, small and medium industries using satellite data.
- Trained a **contrastive learning model** on 7,000+ multi-spectral and hyper-spectral satellite images to infer industrial pollution levels, achieving a 87% accuracy in emission classification across test regions.

### HYUNDAI MOTORS INDIA LTD - Data Analytics Intern

Oct 2024 - Apr 2025

- Conducted **supply chain risk assessment** through real time data analysis and designed interactive dashboards to gather insights.
- Created **audit automation systems** that helped to avoid repetitive work and optimize the project timeline.
- Built a project management system to improve day-to-day team efficiency and to provide actionable insights during meetings

## PROJECTS

### GPT - Architecture MHA Large Language Model

Aug 2026 - Sep 2026

- Built an original Large language model** based in the original GPT architecture from, pre-trained the model and finetuned it for general instruction following.
- The model was build with **my own implementation** of multi-head attention, layer normalization, GELU activation function, 12 layer custom transformer blocks and feed forward implementation

### Communication System for Disaster Relief Operations

May 2024 - June 2024

- Developed a disaster-resilient communication network that transmits requests to relief agencies, when established communication and power systems are down.
- Implemented a **hybrid-search RAG system with re-ranking** for the supervision software that works with the communication system.
- The entire application helps Govt. officials to coordinate rescue and relief operations.

### Iris Tracking Communication Aid for Paralyzed people

Mar 2024 - Mar 2024

- Developed a **OpenCV based Gaze detection** software for fully paralyzed individuals to communicate.
- It tracks eye movements, listens to conversations, and uses AI to generate responses, which are selected by eye movement and spoken aloud.

## KEY ACHIEVEMENTS & CERTIFICATES

- AWS Certified AI Practitioner**
- SAP Hackfest 2024** - 2nd place at national hackathon (**prize: 1.5 lakhs**)
- Smart India Hackathon** - Winner - 2025 (**Prize : 1 Lakh**), Finalist - 2024
- National Hackathon MIT** - Winner (**Prize : 30000/-**)
- IIT PALS INNOWAH National Hackathon** - Winner (**Prize : 15000/-**)
- Delegate** - Invited as delegate and exhibitor at Startup Mahakumbh 2024

## EDUCATION

### B.Tech Artificial Intelligence and Machine Learning

Nov 2022 - Aug 2026

Rajalakshmi Engineering College